ADITYA KADIA

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LinkedIn | GitHub | Portfolio

PROFILE

Computer Engineering (AI & ML) undergraduate with industry experience building software that talks to real hardware and real users. Recent work centres on applied AI: a fully local retrieval-augmented generation (RAG) assistant for HR and labour-law documents, alongside machine learning models for forecasting and prediction. Comfortable across the stack, from Python services and REST APIs to React front ends. Seeking a Master's programme in Computer Science / Artificial Intelligence to deepen work on LLM systems, retrieval and applied machine learning.

EDUCATION

LJ Institute of Engineering and Technology, Ahmedabad *Expected 2027*
Bachelor of Engineering in Computer Engineering (Artificial Intelligence & Machine Learning)

ITM SLS Baroda University, Vadodara *Aug 2021 – May 2024*
Diploma in Computer Science and Engineering | CGPA: **8.05 / 10**

Relevant coursework: Data Structures, Algorithms, Operating Systems, Computer Architecture, Information Security

EXPERIENCE

Software Developer Intern — Payguru Technosoft Pvt. Ltd., Vadodara *Sep 2025 – Mar 2026*

- Built Python applications that connect directly to biometric devices to retrieve attendance records, punch logs and user data
- Developed an automated access-blocking service that restricts system access when an employee's work order expires
- Contributed to development and maintenance of the company's HRMS mobile application in Flutter
- Designed dynamic HR and operational reports in Jaspersoft Studio
- Worked on backend logic, data handling and system integrations supporting internal HRMS workflows

PROJECTS

HRMS-AI: Local RAG Chatbot for HR & Compliance *Python, FastAPI, LangChain, Ollama, ChromaDB/Qdrant, React*

- Built a retrieval-augmented generation assistant that answers HR, payroll, leave and labour-law questions from company documents, running entirely on local infrastructure (Ollama with Qwen3 8B, BAAI bge-small embeddings) so no employee data leaves the network
- Wrote an ingestion pipeline for PDF, DOCX, XLSX, CSV, HTML and TXT with token-aware chunking and per-chunk metadata (company, department, page, section), enabling filtered retrieval and source citations
- Reduced CPU response time from about 7.5 minutes to under 1.5 minutes by suppressing reasoning output, tightening retrieval scope and capping generation length; a relevance threshold answers out-of-scope queries in under 50 ms without calling the LLM
- Added prompt-injection and jailbreak guardrails, JWT authentication with role-based access, streaming responses (SSE), a swappable vector-store layer, Streamlit admin and React front ends, and a pytest suite

GitHub

Aeolus: 24-Hour Wind Forecasting and Turbine Yaw Advice *Python, FastAPI, LightGBM, statsmodels, React, Three.js*

- Built a local web app that downloads three years of hourly wind data for a site, forecasts the next 24 hours of wind speed and direction, and recommends turbine yaw with energy estimates from NREL power curves
- Trained LightGBM against persistence, seasonal-naive and SARIMAX baselines using a time-ordered split, a leakage check and a 596-forecast rolling backtest, cutting wind-speed error by 16.8% and direction error by 10.5%; reported that Open-Meteo's own forecast remained more accurate
- Encoded wind direction as sine and cosine to derive a confidence score, and limited yaw advice to trusted hours with an 8-degree deadband to avoid needless turning
- Delivered a FastAPI backend with background jobs and a React dashboard with a map, charts and a 3D turbine view, covered by 337 tests

GitHub

ApexIQ: F1 Performance Intelligence Dashboard *Python, FastF1, FastAPI, SQLAlchemy, React, Three.js*

- Built a full-stack app that ingests real Formula 1 data with FastF1 into a relational database (re-runnable, no duplicates) and analyses driver and team pace, reliability and teammate gaps
- Forecast races and the championship with a Plackett-Luce Monte Carlo simulation (10,000+ runs) giving win, podium and retirement probabilities; validated models with forward-chaining backtests and rejected those that did not beat the baseline
- Built a Strategy Lab that compares custom and real pit strategies with a tyre-degradation model and an uncertainty band; froze pre-race forecasts so the track record cannot be rewritten

- Reconstructed circuits from position data in a 3D Circuit Lab that replays two-driver telemetry with a lap-time delta; React dashboard covered by 74 frontend tests

GitHub

Automated Security Personnel Post Allocation System *Node.js, Express, Prisma, MySQL, React, Python, Socket.IO*

- Built a real-time platform that assigns security guards to duty posts the moment they punch in on biometric terminals, replacing manual shift rostering
- Developed a multithreaded Python service that polls ZKTeco (TCP) and Hikvision ISAPI (REST) devices, de-duplicates punches and reports device health
- Designed a deterministic two-pass allocation engine: first meets minimum staffing on posts ordered by priority and criticality, then fills buffer capacity, enforcing skill-level and gender rules with a full audit trail
- Delivered a live control-room dashboard over WebSockets with alerts for vacant critical posts and offline devices, email alarms, single-active-session login and role-based module permissions
- Built filterable deployment and attendance reports with CSV export for HR and payroll, and a simulation mode that generates live punches so the full pipeline can be tested without physical hardware

GitHub | Live demo

Smart Inventory: Multi-Tenant Inventory SaaS with ML Forecasting *React, Node.js, Express, MariaDB, FastAPI, scikit-learn*

- Built a multi-tenant SaaS for businesses tracking stock across shops and warehouses, with a separate database per tenant, request-level tenant routing, cached connection pools and automated tenant provisioning from a Super Admin portal
- Implemented a stock movement ledger (stock-in, stock-out, adjustments, inter-location transfers) with a full audit trail, and a purchase-order workflow from draft through approval to partial or full receipt
- Developed a Python FastAPI ML service: linear regression for stock-out timing, Holt-Winters exponential smoothing for 7/14/30-day demand forecasts, Isolation Forest for dead-stock detection, K-Means to segment fast, medium and slow movers, and EOQ-based reorder quantities
- Added bulk Excel/CSV catalogue import with row-level validation, staff invitations with role-based access control, bcrypt-hashed passwords and a background email queue

GitHub | Live demo

Expense Tracker with Automated Bank-Email Sync *React, Node.js, Express, MySQL (TiDB), Sequelize*

- Built a full-stack personal finance app supporting multiple bank accounts, categorised income and expenses, and Excel export
- Automated transaction capture by parsing bank alert emails (HDFC, ICICI, SBI) through the Gmail API on an hourly scheduled job; detected transactions go to a review queue to approve, edit or dismiss
- Secured Gmail access with read-only Google OAuth2 scopes and AES-256 encryption of stored tokens, alongside JWT login and bcrypt password hashing
- Built dashboards with Recharts showing balance, 30-day expense and 60-day income trends; deployed frontend and serverless backend on Vercel

GitHub | Live demo

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, Dart
AI / ML: scikit-learn, Pandas, NumPy, LangChain, Ollama, Hugging Face Sentence-Transformers, RAG, prompt engineering, LightGBM, statsmodels, SciPy, time-series forecasting
Backend: FastAPI, Django, Node.js, Express, REST APIs, WebSockets (Socket.IO), JWT / RBAC, Pydantic, SQLAlchemy, FastF1, Open-Meteo API
Frontend & Mobile: React, Vite, Material UI, Tailwind CSS, HTML, CSS, Three.js, React Three Fiber, Recharts, TanStack Query, Leaflet, Anime.js, Streamlit, Flutter
Databases: MySQL, MariaDB, MongoDB, SQLite, Firebase, ChromaDB, Qdrant, Prisma ORM
Tools: Git, Docker, Jaspersoft Studio, pytest, Vitest, ESLint

CERTIFICATIONS

Building Generative AI-Powered Applications with Python — IBM (Coursera) Verify *Jul 2026*
Introduction to Software Engineering — IBM (Coursera) Verify *Mar 2026*
Back-End Development with .NET — Microsoft (Coursera) Verify *Mar 2026*
ReactJS — Scaler Topics *Jul 2025*
Exploratory Data Analysis for Machine Learning — IBM (Coursera) Verify *Jun 2025*
Introduction to HTML, CSS & JavaScript — IBM (Coursera) Verify *Dec 2024*